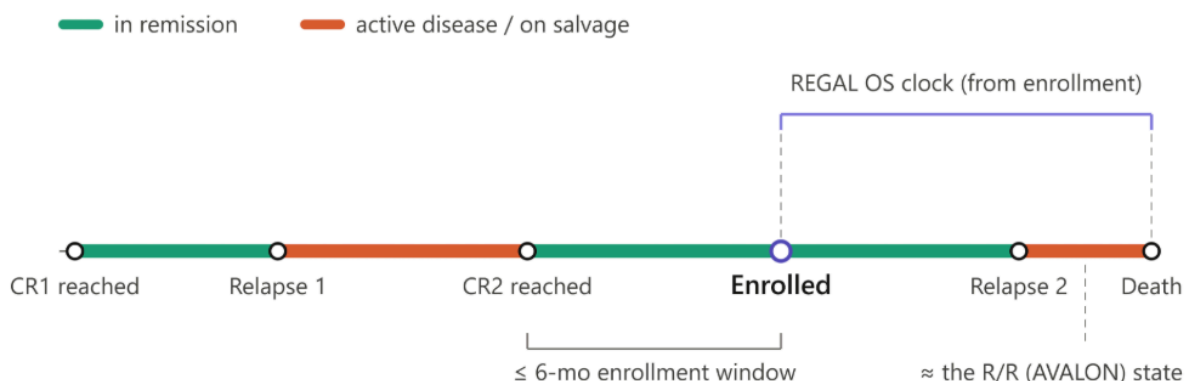


1) Clarifying where REGAL patients actually sit

The most common error I see is literature comparisons that conflate the **CR2** population with the **relapsed/refractory** (R/R) population. Those data get cited frequently to insist that BAT median OS "can't" exceed 6-8 months. That inference is wrong, and rests on a misunderstanding of who is in the trial.

The intuition behind the error is that CR2 comes *after* R/R on the disease timeline, so CR2 survival must be *worse*. I have seen this same error repeated many times at other points in the AML trajectory such as comparing CR1 to newly diagnosed as well. However, survival in a disease state is not determined by where it sits on a timeline but rather is determined by **who survives to occupy that disease state**. The R/R population is a blend. It includes patients who relapse and die quickly without ever getting back into remission. These salvage failures dominate the pooled curve and drag the median down. REGAL does not enroll them, it enrolls the selected subgroup that has already cleared the single hardest filter in the disease: **getting back into remission**.



The problem is that almost no published data measures OS from second CR specifically. So we have to reconstruct it from adjacent populations. That is what the rest of this analysis focuses on.

I have previously written that the **3-yr OS in R/R AML is effectively zero without a bridge to transplant**. I got substantial pushback, with people citing 5-15% three-year survival without transplant. I want to restate the claim precisely since a loose version invited objection:

Conditional on being in active relapsed disease (R/R AML), long-term survival without transplant, and durable CR3 achievement, are close to zero

The data support this. To name just a few:

- Liew-Littorin (2025): 4 of 93 primary refractory patients reached 3 years without transplant (~4%) AND primary refractory disease is a population with *better* prognostics than relapsed disease
- AVALON: 9 of 147 patients alive at 2 years, 6 of whom had been transplanted, leaving only 3 patients or ~2.8% of the non-transplanted cohort alive at 2 years.

Low single digits. "Effectively zero" is a fair description of that.

That 5-15% figure people quote isn't wrong however but is **measuring from a different bucket**. Once you accept that active relapsed disease has essentially no long-term survivors without transplant, the population divides cleanly into three:

1. **Those who reach remission and stay there** (favorable biology, durable CR2). This is where the 5-15% comes from.
2. **Those who relapse again and die**
3. **Those who bridge to transplant**

Remember these buckets as they will be relevant for how we model survival later in this analysis.

2) A first pass attempt at reconstructing the control from summary statistics

The natural first attempt is to decompose the object we want:

OS from enrollment $\approx$ (enrollment $\rightarrow$ second relapse) + (second relapse $\rightarrow$ death)

Call these the **remission segment** and the **active-disease segment**.

The **active-disease segment** is covered well by the modern R/R literature:

Source	Median OS
Bataller	5.3 mo
AVALON	6.3 mo

R/R AML on VEN+HMA	5.5 mo
Gilteritinib (subgroup)	9.3 mo
Ivosidenib (subgroup)	9.0 mo
Enasidenib (subgroup)	9.3 mo

The targeted-agent figures describe biomarker-selected subgroups and don't generalize. What they show (usefully) is that the aggregate patient curve is fairly robust to this class of therapy. A modest subgroup with somewhat better survival will not reshape the pooled curve substantially.

Furthermore, since these studies include many patients in *first* relapse, survival in *second* relapse is likely somewhat lower

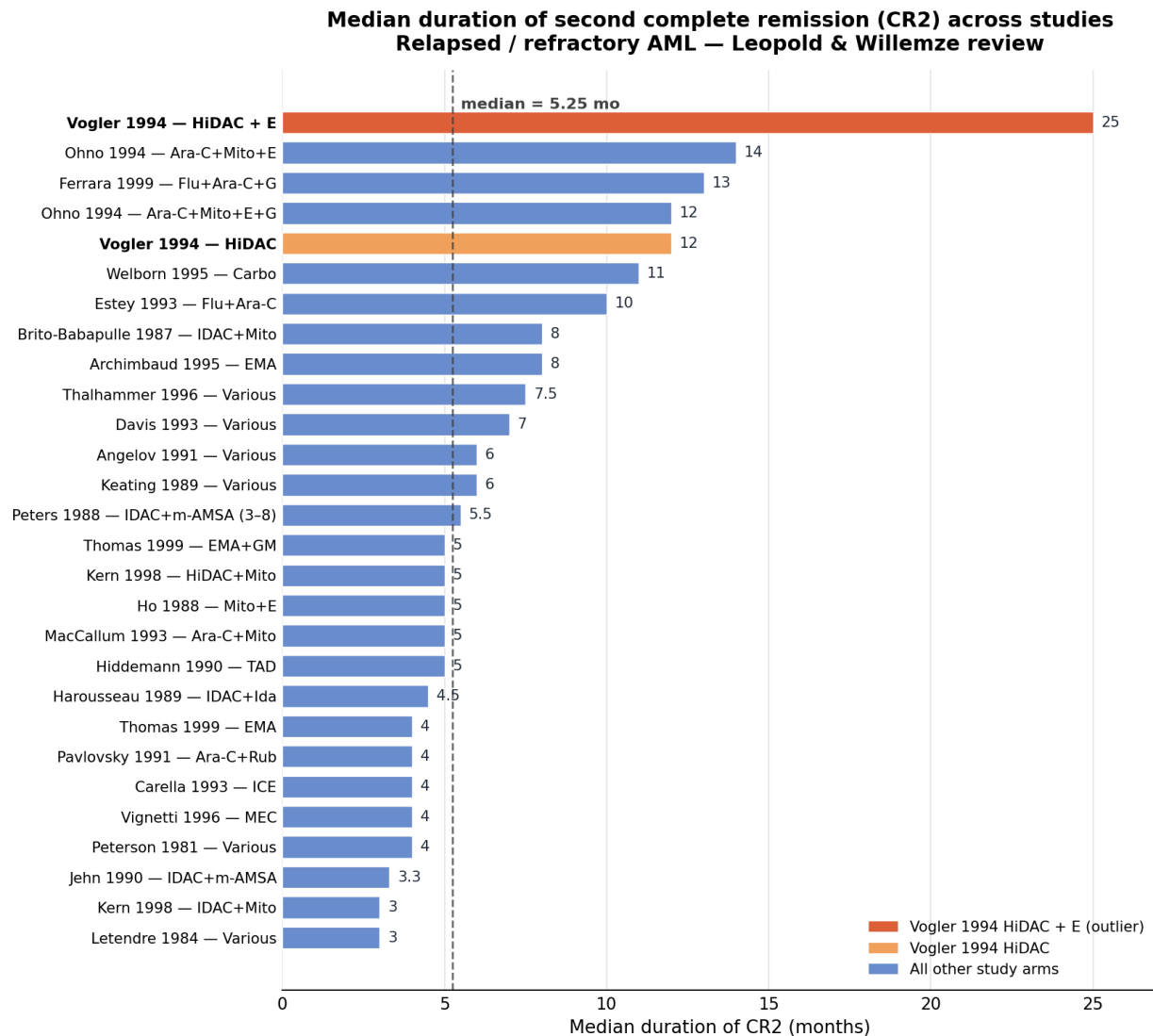
≈ 5-6 months.

The **remission segment (duration of CR2)** requires us to cast our net further back in time where I was able to surface the following datapoints:

- **Thalhammer (1996)** — 7.5 months, the most widely cited datapoint
- **Davis (1993)** — 7 months, for patients who did *not* proceed to intensive consolidation with BMT
- **2001–2006 cohort, ages 50–70** — 8.5 months
- **Northern Italy (2008)** — 3–4 months (range 1–34), small high-risk cohort on intensive chemo alone
- **Uhlman (1990)** — 5 months
- **Netherlands (1990)**, poor-prognosis R/R — 8.2 months
- **Idarubicin + intermediate-dose cytarabine (1989)** — ~4.5 months

Most valuable was **Leopold & Willemze (2002)**, a comprehensive review reporting median CR2 duration across 28 study arms by treatment combination. The range was **3-14** months, with one extreme outlier at 25 months that both I and the study authors exclude (it wasn't statistically

significant, and only the abstract was ever published, so there's no KM curve to assess how fragile it was).



also want to flag the enasidenib R/R trial, where median response duration was 8.8 months among patients achieving CR (i.e. CR2 duration), with median OS of 19.7 months in that group. It is a narrow bio-marker subgroup so not broadly applicable, but it is encouraging that it lands in the same range. In summary, CR2 duration

≈ 4-10 months.

3) Does the modern era push CR2 duration up?

Since nearly all of the CR2-duration data is old (for some reason it fell out of vogue to report this), we have to reason from first principles about what the changing treatment landscape has done to CR2 duration. The naive reasoning — *better drugs, therefore longer remissions* — does not necessarily hold.

The key point is that you cannot just ask whether a drug is better. You have to ask **who the drug moves in and out of the CR2 pool**, because CR2 duration is a property of the pool's composition. A therapy that helps every single patient it touches can still make the measured CR2-duration statistic *go down*, if it changes who is being measured.

Two important things have changed in the treatment landscape.

Force 1 is drug innovation. For approximately fifty years, 7+3 was the dominant standard of care. Recently we've seen the "sunset of 7+3" and the sunrise of hypomethylating agents with BCL-2 inhibition, plus targeted agents against FLT3, IDH1, IDH2, and NPM1.

Force 2 is transplant innovation. In the last decade, a new technique known as post-transplant cyclophosphamide has rewritten who can be transplanted.

One framing note before I start, there are two different objects in play here that are affected in two different ways:

- (A) **The CR2-duration term** — which measures how long a modern no-transplant second CR lasts.
- (B) **REGAL's BAT arm** — the actual OS of the actual patients in the actual control arm.

These are not the same thing, and I'm going to argue that the modern era pushes **(A) down** and **(B) up**. That sounds contradictory but it isn't and will be elucidated in this section.

Force 1 — The Drug Cascade

Lengthening channel (weak, and protocol-excluded)

A patient who would have reached CR2 anyway now has a targeted agent that extends their maintenance. This is the mechanism that would push CR2 duration up. However, it's thin:

- **HMA + venetoclax produces shallow remissions** (typically CRi/CRp rather than full CR).
- **Venetoclax + oral azacitidine maintenance failed for futility in VIALE-M**
- That leaves **molecularly-targeted agents which the European protocol tells us REGAL excludes**: "*patients whose remission in CR2 can be maintained with molecularly targeted agents (e.g. FLT-3 or IDH inhibitors) per investigator's determination.*" This class of drug is not one that tends to get "toggled" while the patient sits in remission based on changing facts since the decision rests mostly on a molecular characterization that determines the targeted-agent decision flows.

So the one channel that would plausibly lengthen CR2 duration largely fails.

Shortening channel

The new agent classes **expand who can achieve CR2 at all**, and that expansion disproportionately admits biologically higher-risk, less-fit patients whose remissions are short. Before the targeted era, a patient refractory to 7+3, or an older patient who couldn't tolerate intensive salvage, **simply never achieved CR2**. They contributed **zero** entries to the CR2-duration distribution since they died in active disease and landed in the OS-from-first-relapse numbers instead.

Today: gilteritinib coaxes FLT3-ITD patients into remission. HMA + venetoclax coaxes the unfit older patient in. An IDH inhibitor coaxes the chemo-ineligible IDH-mutant patient in. **Most of these newly-rescued patients enters the CR2 pool at its left tail** with short, often shallow remissions since these are cases where high-risk biology is rescued by *disease-controlling* rather than curative drugs.

This actually likely exhibits a **Will Rogers phenomenon** where measured outcomes get worse in both the pre-remission and post-remission cohorts even though every individual patient is being treated better. This happens because you are *moving the best of the R/R patients into the worst of the CR2 group*. R/R survival falls AND CR2-conditional duration falls, because the CR2 pool gains fragile remissions it never used to contain.

Transplant-skimming channel

Allo-HSCT is the only genuinely curative treatment option in these patients (with roughly 30-50% long-term survival amongst patients who receive one). It is preferentially undertaken by patients who are both fit enough and responding deeply enough to survive it. Therefore, the key question is how these modern agents have changed the supply of qualifying remissions (i.e. acting as a bridge to transplant).

Whether a newly-rescued patient actually depresses the no-transplant pool depends entirely on whether they are then skimmed straight back out of it. We can trace each class the whole way through:

rescued into CR2 by	remission quality	then extracted to transplant?	net effect on the no-transplant pool
Gilteritinib (FLT3-ITD)	fragile, fast-relapsing	yes — near-completely. Everyone knows it won't hold	≈ nil (passes through)

Menin inhibitors (KMT2A / NPM1)	4.5–5 mo DoR	yes — urgently	≈ nil (passes through)
HMA + venetoclax	shallow (CRi, MRD+)	no (the comorbidity that makes the remission shallow is the same comorbidity that makes them poor transplant candidates)	DOWN — this is the one that does the damage
IDH inhibitors	mixed	partial	UP — see below

So the drug cascade's downward pressure on the no-transplant CR2 term is carried almost entirely by HMA + venetoclax. Gilteritinib and the menin inhibitors seed the CR2 pool and then leave it.

IDH inhibitors split their responders along a prognostic line which makes them two-sided. Responders whose remission clearly won't hold (adverse co-mutations, shallow depth) get bridged but responders with more favorable biology can achieve a durable CR and **simply stay on the pill**. Patients doing fine on a well-tolerated, once-daily oral drug that has a known long-tail aren't going to be urgently bridged. Extraction in the IDH channel is therefore partial but it is depositing good patients into a no-transplant pool so is a real upward force on CR2-duration.

However, we can go back to the REGAL exclusion criterion: *"patients whose remission in CR2 can be maintained with molecularly targeted agents (e.g. FLT-3 or IDH inhibitors)."* So the durable patients on IDH inhibitors will be excluded and what is left is the slice that responded but not durably, and wasn't transplantable.

Force 2 — The Transplant Cascade

In 2008, a Johns Hopkins group published the landmark work showing that **post-transplant cyclophosphamide (PTCy)** made haploidentical transplant feasible in AML, with a dramatic reduction in severe graft-versus-host disease. Major centers spent the following years validating it, and by roughly 2014-2017 there was a worldwide inflection: haplo went from a last-resort procedure that was rarely used because of its toxicity to a standard option.

PTCy achieved two separate things.

It removed the donor barrier. Before PTCy, if you had no matched sibling and no matched unrelated donor (MUD), you simply did not get an allograft regardless of fitness or how deep your remission was. Donor availability was a much larger bar that had nothing to do with your disease (it was also considerably worse for patients with non-European ancestry whose registry match rates were far lower). Haplo donors, by contrast, are closer to universal.

It lowered the toxicity ceiling. By cutting severe GvHD, PTCy cut non-relapse mortality, which lowered the fitness bar for transplant. Combined with reduced-intensity conditioning, patients who a decade earlier simply could not have survived an allograft became transplantable.

Now, I want to be careful for this next part because the obvious reading and the correct reading are not so transparent.

Before enrollment: PTCy skims the pool

The donor channel removes the patient who was **fit, in a deep remission, and simply unlucky**. That patient was sitting in the historical no-transplant CR2 cohorts propping their curves up from the top. Removing that patient lowers the residual.

The toxicity channel *looks* like it should cut the other way, since it removes *unfit* patients, who are bad-prognosis. It doesn't, and the reason is: **transplant is never offered at random**. You still need a deep enough remission and good enough performance status to justify the procedure. So when PTCy pulls previously-untransplantable patients across the line, it pulls **the best-remission, best-performing subset of the unfit** — not a random draw of them, and certainly not the worst. This is another Will Rogers phenomenon in its own right. Therefore, the organizing principle is:

*Every extraction to transplant is positively selected. Any mechanism that increases extraction (more remissions, more donors, less toxicity) removes patients who were **above the pool's average**.*

Thus, on the pre-enrollment leg, PTCy makes the modern untransplanted CR2 population a more adversely selected residual than any historical cohort. Therefore, we expect **object (A)** to decline.

After enrollment: PTCy makes REGAL's barrier leak

Since REGAL does not enroll a *permanently* untransplantable population. It requires ineligibility only at the moment of randomization. PTCy helps turn transplant-ineligibility into a more **transient** state.

Consider what "transplant-ineligible" meant, then vs. now:

- **1995:** No matched donor → *likely permanent*. Too frail for myeloablative conditioning → *likely permanent*. These were terminal classifications. You were never getting a transplant, ever (most likely).

- **Today:** Everyone can get a haplo donor, and RIC + PTCy dropped the fitness bar dramatically. So what's left as "ineligible" is things like *active infection, counts not yet recovered, comorbidity being optimized, MRD not yet deep enough, patient hasn't decided*. **These are reversible conditions.**

PTCy did not just skim the pool. It made the wall that defines REGAL's control arm a leaky one. That arm will bleed patients into transplant across follow-up

Therefore, we expect **object (B)** to have the opposite sign and the comparators to understate REGAL's control arm (rather than overstate).

The Takeaway

(A) The CR2-duration term: flat-to-lower.

Relative to the 1990s salvage trials, this logic chain implies **modern no-transplant CR2 duration should be flat-to-lower, not higher, despite the drugs being unambiguously better.**

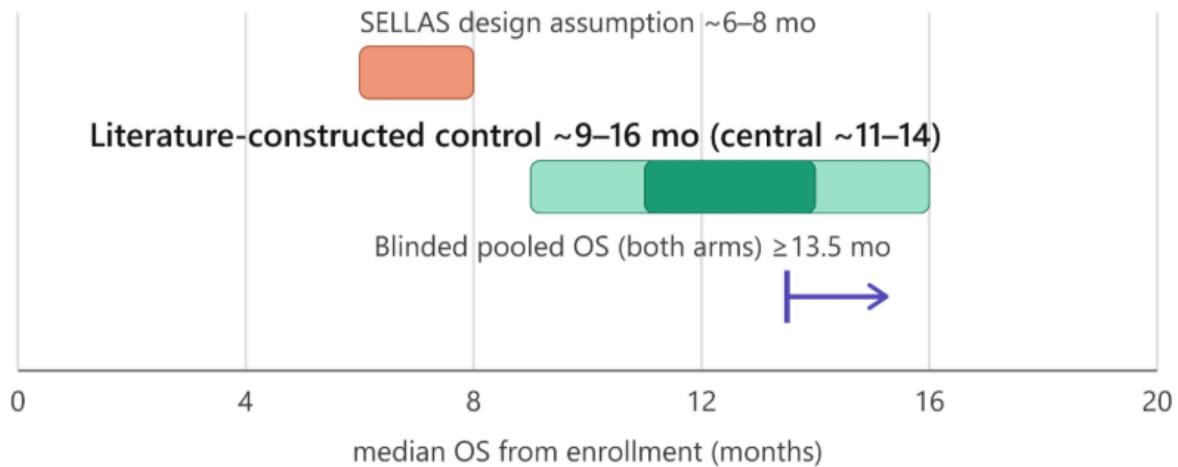
The older data measured a CR2 pool reachable almost exclusively through intensive salvage chemotherapy which was a fitter, more favorably-selected population that was less aggressively skimmed of its durable responders. Today, HMA+ven seeds the left tail with unbridgeable fragile remissions that older cohorts never contained. Similarly, the skimming channels extract the durable right tail more.

(B) REGAL's BAT arm: higher than any never-transplant reconstruction implies.

We've established REGAL's ineligibility condition is a snapshot. Aside from that, however, as "no donor" and "too toxic" evaporate as reasons for skipping transplant, **refusal and preference become a larger share of what's left in the ineligible pool.** Refusers are not adversely selected and some are **favorably selected** (e.g. the core-binding-factor patient who declines an allograft precisely because their odds of durable remission are already good without one). As the involuntary reasons disappear, the voluntary ones that make up the remainder bring better biology with them.

4) Why you cannot just add the R/R median and the CR2 duration median

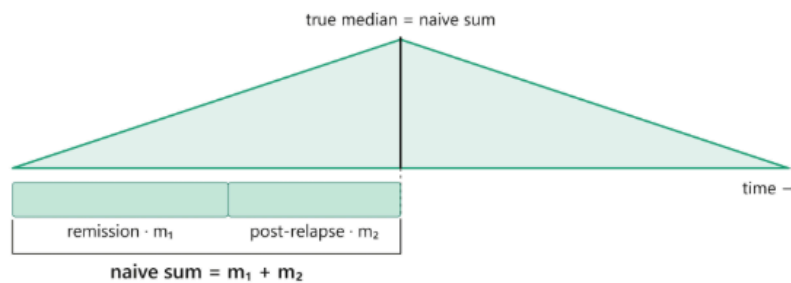
Now that we have analyzed the applicability of the older data to the modern era, we must address another problem. Combining the two pieces gives a first-pass literature-constructed control: a **~4–10 month remission segment plus a ~5–6 month active segment.**



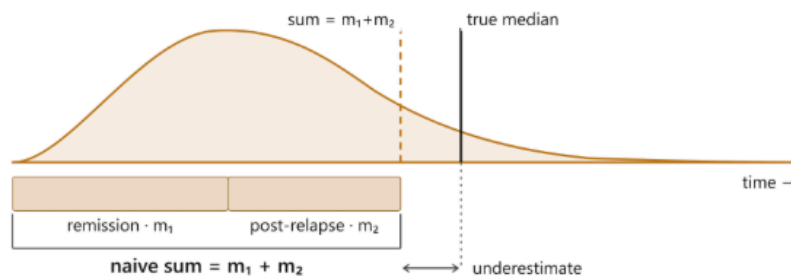
However, this is only valid as a grounding exercise because **naive segment summing is not a valid way to estimate a blended median**.

Adding medians only reproduces the true median when each segment's event-time distribution is roughly **symmetric**. Survival distributions are not. They are right-skewed.

Phases linear (symmetric) — naive sum is exact



Phases exponential (skewed) — naive sum underestimates



Two additional factors interplay:

- **The segments are almost certainly correlated.** Adverse biology tends to produce both a short CR2 and shorter post-relapse survival. In the limiting case of perfect rank-correlation, medians add *exactly* and the underestimate vanishes. The true

correction is somewhere between substantial and zero, and summary statistics can't pin it down.

- **The decomposition has no room for death in remission.** Some patients die in CR2 without relapsing. A two-segment model can't see them.

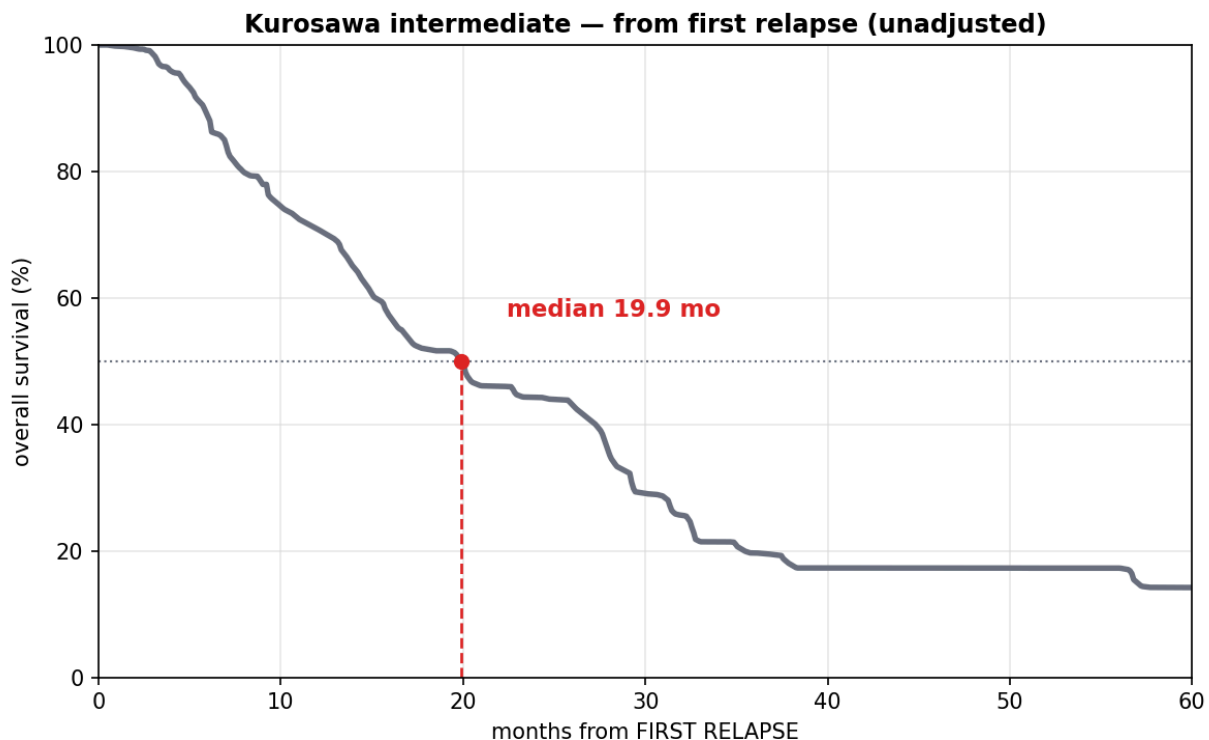
Which brings us to the essence of the point: **with point estimates drawn from different populations, you cannot recover a true median at all. You need a full curve.**

5) The right approach: a matched KM curve

Rather than inferring from summary statistics, a more accurate comparator is to take a full KM-curve from a single patient population and transform it into a REGAL-aligned object. I found three studies where this is feasible: **Kurosawa**, **Burnett**, and the supplementary data for **Van der Maas**.

5a. Kurosawa

I digitized the KM curve for the **intermediate-risk, CR2, no-HCT** cohort.



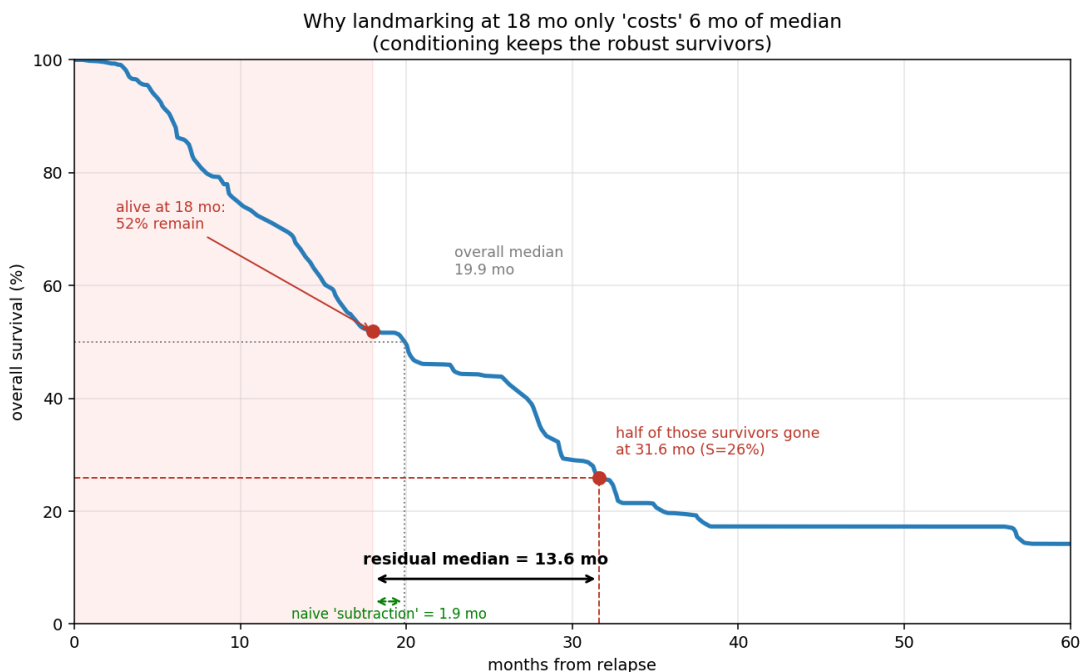
Median OS is approximately **20 months**. This cohort resembles REGAL in an important way: conditioning on the same selection event of actually achieving CR2. However, two adjustments are required.

Adjustment 1: removing the pre-CR2 stretch. Kurosawa measures from the onset of post-CR1 relapse even though it conditions on actually achieving CR2. Therefore, it spans *relapse* → *CR2* → *death*. We only want *CR2* → *death*. Ideally, we would subtract time-in-relapse per patient, but that requires individual patient data (IPD). Fortunately, among patients who reach CR2, median-time-to-CR/median-time-to-best-response is short and tightly distributed (typically **1-3 months** across a wide range of studies). So we approximate by subtracting two months from every patient, **shifting the curve two months to the left along the time axis**. This doesn't bias the result, because it conditions in the same way REGAL does.

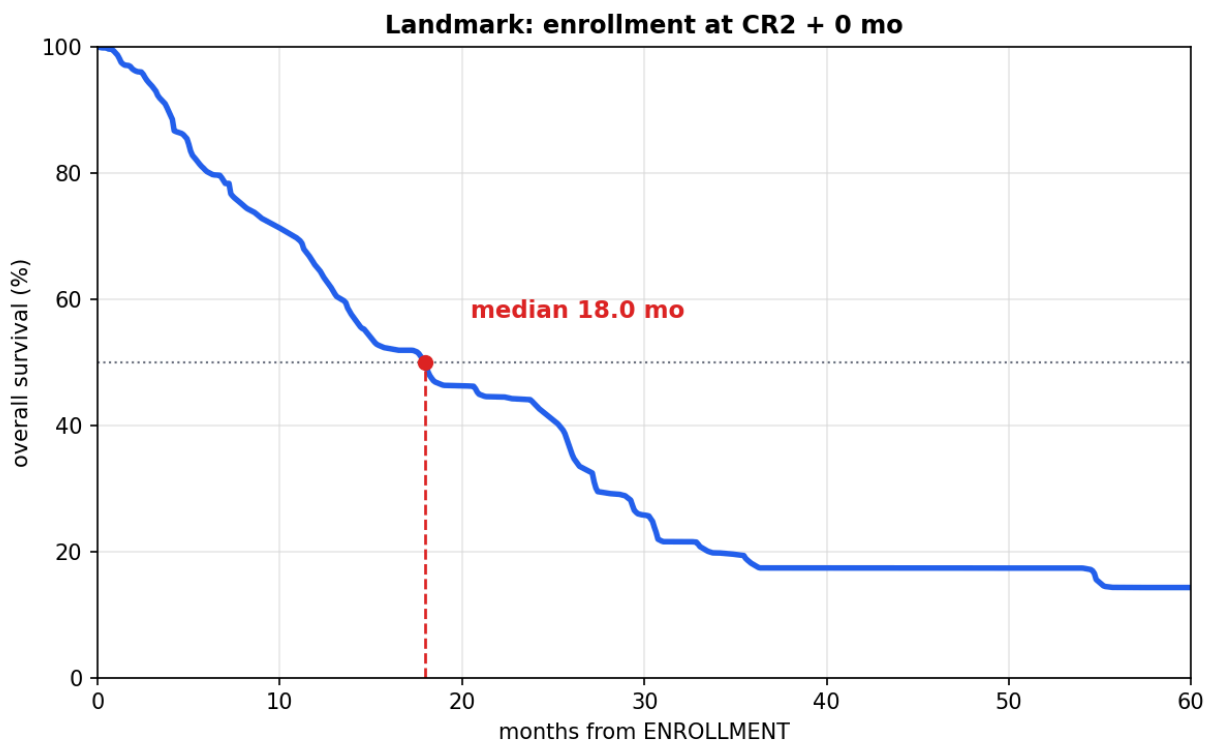
Adjustment 2: REGAL's enrollment window. Patient's are not enrolled the instant they hit CR2; there's a window of up to ~6 months. This is **not** a simple subtraction because the window doesn't preserve population comparability but rather applies a **selection effect**. Anyone who dies between CR2 and enrollment never makes it into the trial. This is the classic problem in comparing a retrospective cohort to a prospective trial: **delayed entry bias**.

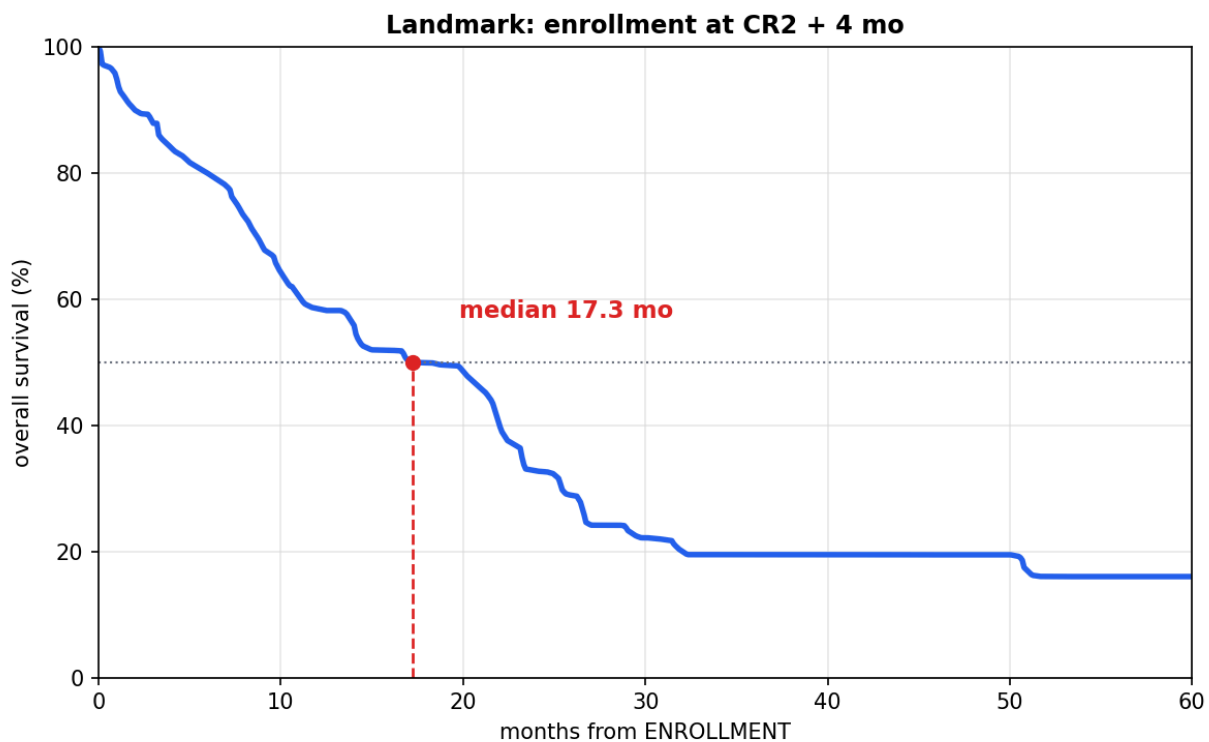
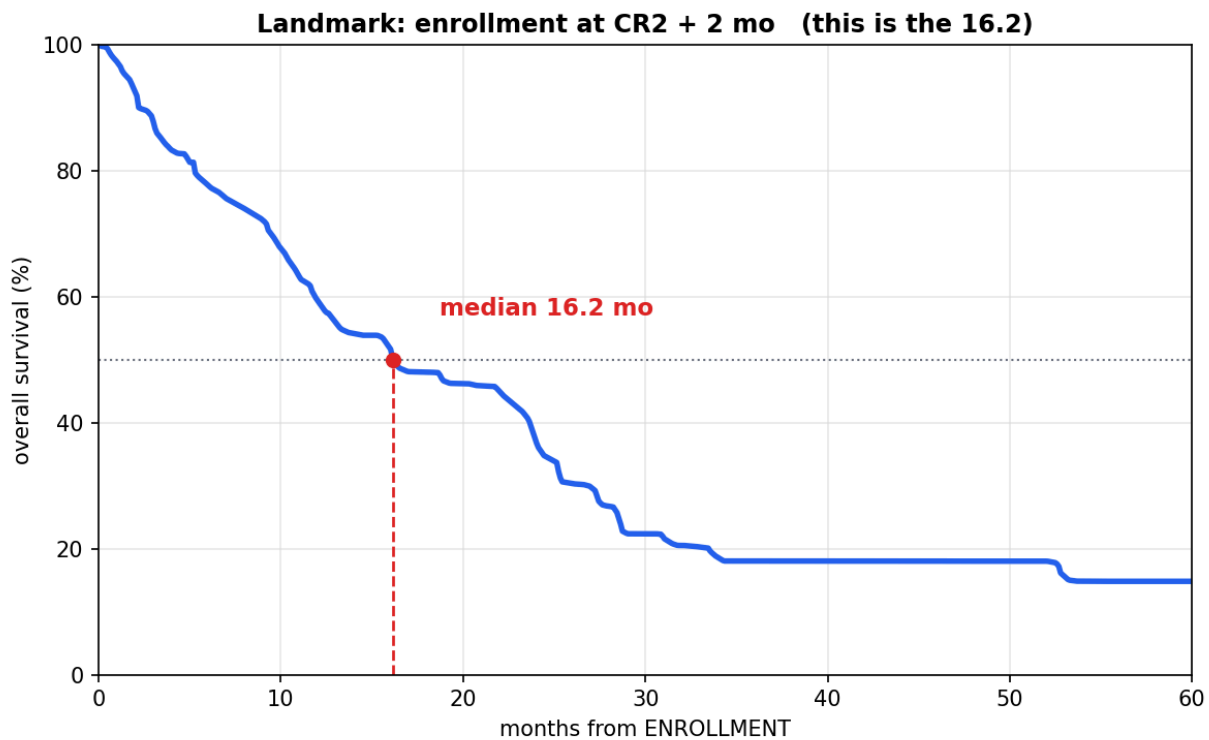
Thankfully, tools exist to address this. We can apply a technique known as the **landmark method** to make a good first-order approximation of this bias (although it is not statistically exact since hazards aren't "memoryless"). The basic concept can be thought of as "sliding down" the curve. If the landmark is 3 months, we drop everyone who died before month 3 and treat survival thereafter as a fresh cohort starting there.

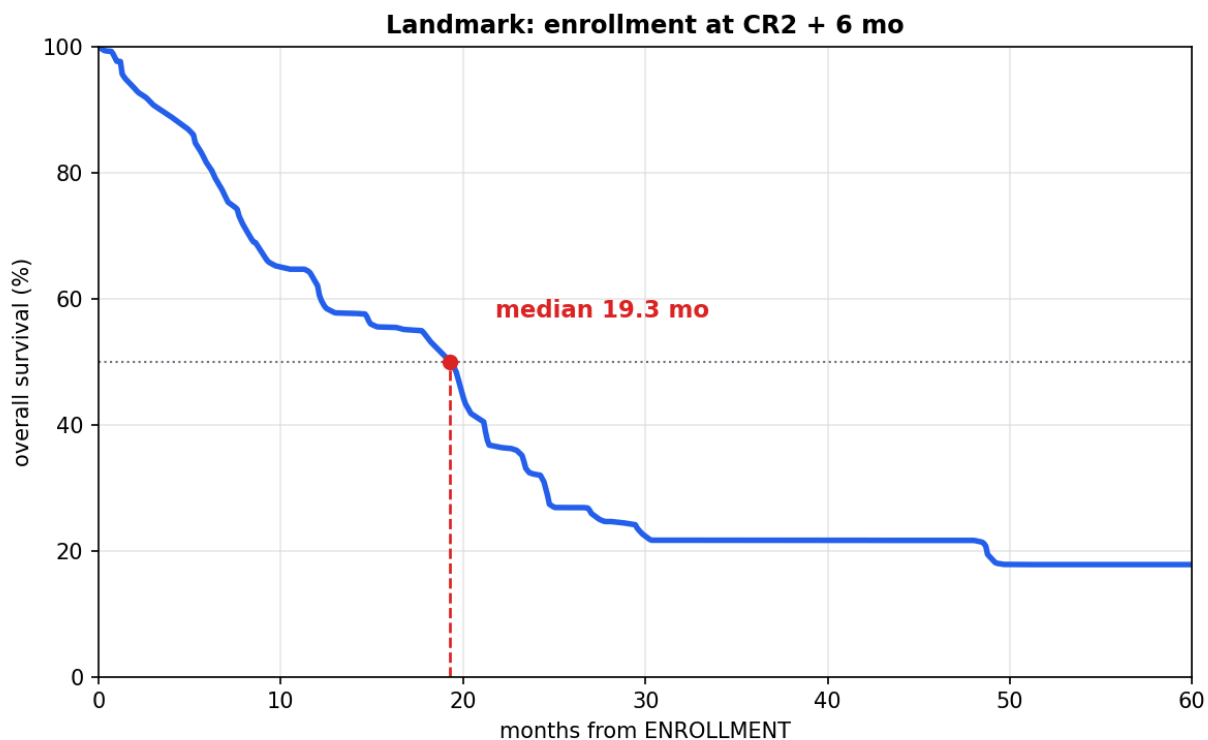
Crucially, you do not simply subtract the landmark time from the median. To see why, here's a hypothetical landmark at 18 months: the residual median is **~13.6 months** not 1.9 months. Conditioning on survival to the landmark keeps the robust survivors, and they carry the curve a long way.



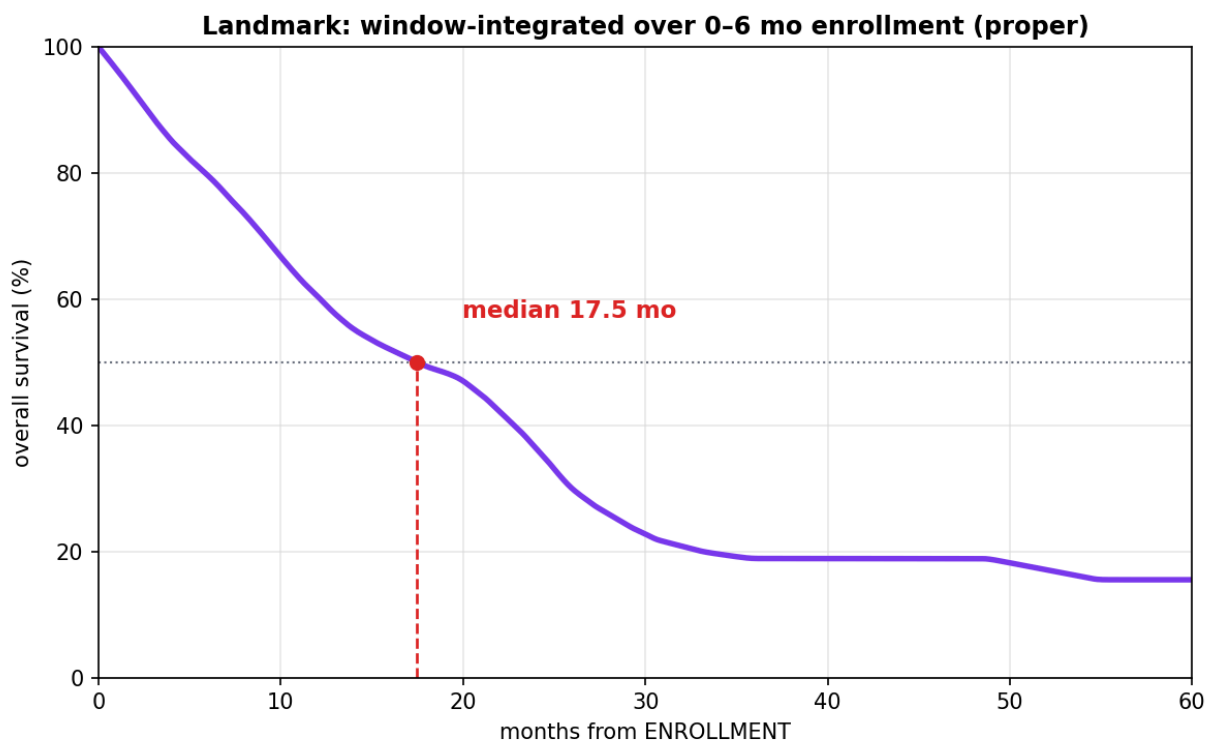
A single "most reasonable" landmark of 2 months gives a landmarked median of **16.2 months**.

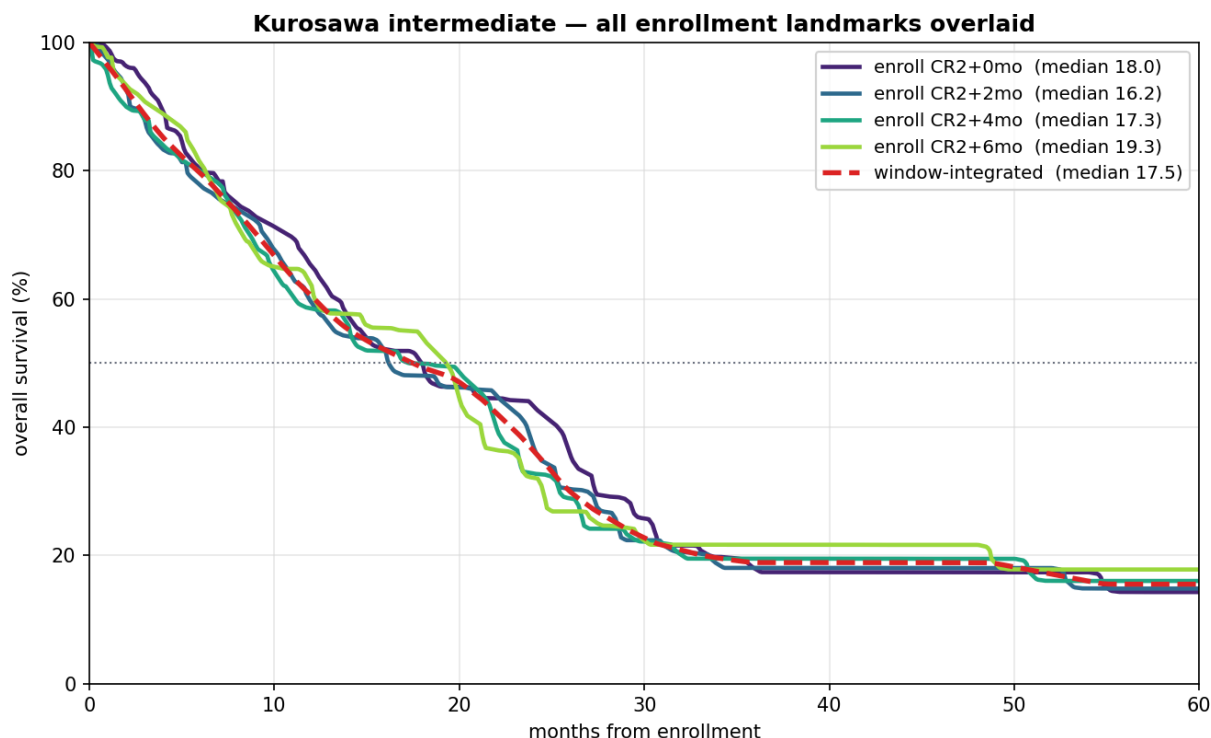






However, we don't have a single landmark — we have a **distribution of entry times**. So, the approach I took is to integrate the landmark adjustment across the entire enrollment window, approximating the full mixture-over-entry-time survival transformation.





Window-integrated median: ~17.5 months. The overlaid landmark curves cluster tightly, so the result isn't hypersensitive to the entry distribution.

One assumption I want to emphasize is that I model entry times as **uniform** on $[0,6]$. Real enrollment is almost certainly **front-loaded** since screening, consent, and logistics would cluster near CR2 confirmation. A front-loaded entry distribution pulls the result toward the CR2+0, and CR2+2 curves, so this is a real (if modest) dependency.

REAL OBJECT (Target: delayed-entry / left-truncated survival)	MY APPROXIMATION (Uniform-landmark mixture)
$S_{\text{REGAL}}(t) = \int_0^\tau S(t \mid T > r) f_R(r) dr$	$\tilde{S}(t) = \frac{1}{\tau} \int_0^\tau S(t \mid T > r) dr$
Where: $S(t \mid T > r) = \frac{S(t)}{S(r)}$ for $t \geq r$ $f_R(r)$ = density of CR2-to-randomization (entry) times τ = maximum possible entry time (study window)	Where: - Same $S(t \mid T > r) = \frac{S(t)}{S(r)}$ for $t \geq r$ - Assumes $f_R(r) = \frac{1}{\tau}$ for $0 \leq r \leq \tau$ τ = maximum possible entry time (study window)
Interpretation This averages the conditional survival over the actual distribution of randomization times. Each landmark r is weighted by how often patients actually entered the study at time r .	Interpretation This averages the conditional survival over all landmarks equally. It is equivalent to assuming that entry times are uniformly distributed on $[0, \tau]$.

Constraints on inferences from Kurosawa:

- **Population.** Data was collected only at Japanese centers, across 1999-2006 which is a period with a different treatment-selection cascade. The geographic difference is partially mitigated by using the **intermediate-risk** curve specifically, which controls for different cytogenetic-risk distributions across ethnic populations (notably the higher reported frequency of t(8;21) in Japanese patients).
- **Transplant eligibility.** Kurosawa's "no-HCT" group is defined **retrospectively** — patients who happened not to receive a transplant, for any reason: no donor, patient declined, center practice, or unfitness. REGAL's BAT arm is defined **prospectively** as patients ineligible for or unable to undergo allo-HSCT, which in practice means comorbidity and lower fitness but also transplant leakage and selection for favorable cytogenetics. My timing adjustments do nothing for this, and it cuts both ways.
- **Tail instability.** The intermediate-risk curve is **n=82**. When digitized, the final vertical drop runs from roughly 18% to 14.5% survival which is a 20% relative drop, implying 5 patients became 4. However, ~18% survival in n=82 should represent 14-15 patients, not 5. **The tail is being carried by a handful of patients under heavy censoring, which makes the 3-yr OS estimate of ~18-20% entirely unreliable.** The censor marks are visible but too grainy to bound the range (i.e. to compute what 3-year OS would be if every censored patient died immediately versus survived long-term). This also implies that our computation for the **median itself** might be sensitive and carries a wide confidence interval.

5b. Burnett

Burnett has the distinction of being the one study I could find in the entire literature that measures the exact object we want: **OS from second CR, in patients without prior SCT**. Unfortunately several things make it an unclear comparator.

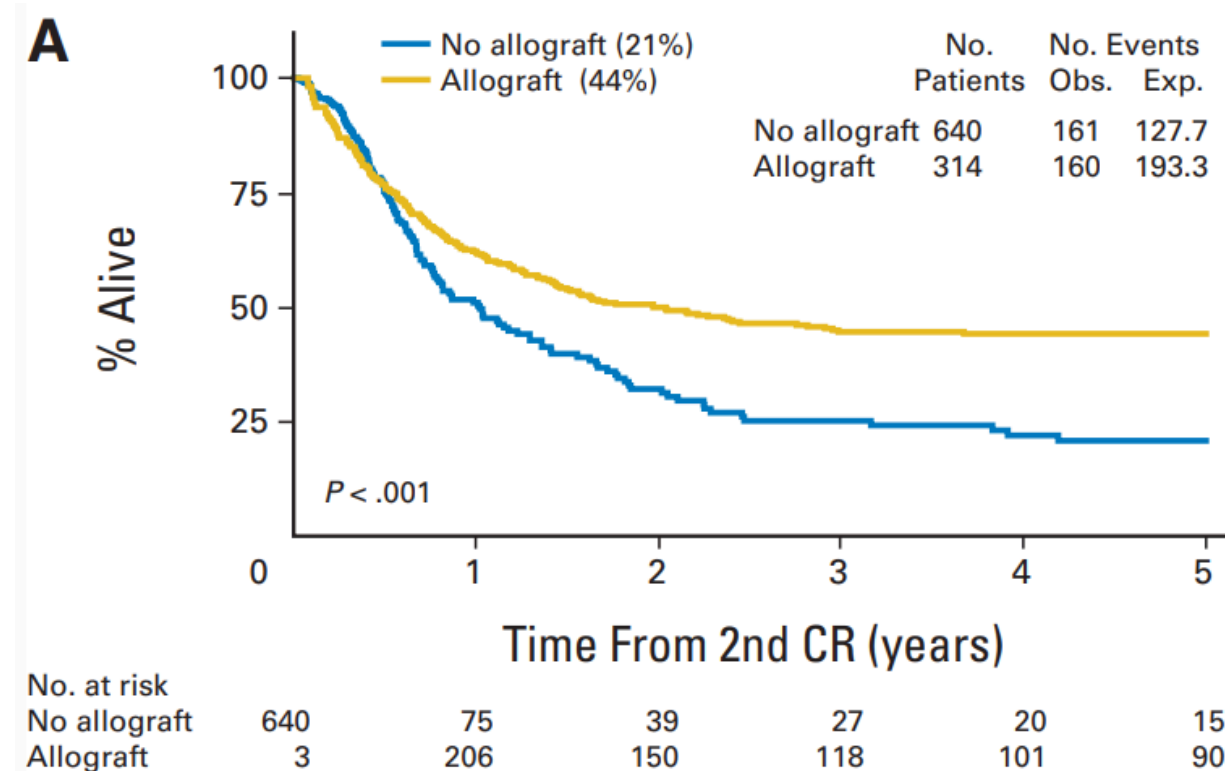
Age. Three MRC trials recruiting 1988-2009, ages 16-49 constitute Burnett. The CR2 group was 28% aged 16-29, 30% aged 30-39, and 42% aged 40-49, median age 37. The obvious assumption many make is that a younger cohort must be fitter, so the REGAL curve should sit *below* this one.

That does not follow, and may in fact invert. Younger patients are enriched for favorable risk (inv(16), t(8;21), NPM1-mutated without high FLT3-ITD), while older patients are enriched for complex and monosomal karyotypes. Young patients also have organ reserve and access to intensive therapy in CR1, so a young patient who achieves CR1 has a relatively high chance of bridging to transplant or simply being cured and never relapsing. That creates a selection effect: **the young patients who nevertheless relapse and get re-induced into CR2 are the ones whose disease was aggressive enough to relapse despite every advantage (which might be a marker of bad underlying biology).**

Put plainly, here is the inversion: in a 25-year-old, skipping transplant is likely dominated by *"the disease moved too fast."* In a 65-year-old, it's likely dominated by *"age and comorbidity."* **Youth doesn't enrich the no-allograft group for survivors — it strips out benign reasons for not transplanting and leaves only the aggressive-biology reason.** The younger and fitter the base cohort, the more adverse the residual that fails to reach transplant. Burnett's ~12 months could easily be low **because** the cohort is young, not despite it.

Transplant era. Every transplant in the study is a sibling allograft, matched unrelated donor, autograft, or reduced-intensity transplant. There are **no haploidentical transplants at all.** The 2009 recruitment cutoff sits right on top of the PTCy inflection described in Section 3, which means Burnett's no-transplant group is rationed by donor availability and by toxicity in a way that no contemporary cohort is. It is the cleanest illustration of the Force 2 argument: **its no-transplant arm still contains all the fit, deeply-remitted, unlucky patients that a modern no-transplant arm would have long since had haplo-grafted out of it.** Whatever Burnett's no-transplant curve is measuring, it is not who ends up untransplanted today.

The statistics. The curve does not track two fixed cohorts. It's a **Mantel-Byar analysis:** all patients start in the no-allograft arm and are switched to the allograft arm at the moment of transplant, being censored from the original arm at that point, with survival estimated by the **Simon-Makuch** estimator. Mantel-Byar is the gold standard for eliminating immortal-time bias, but it produces curves that are much harder to read, because the effective sample size and the *composition* of each group change continuously over time.



In this study, the effective sample size collapses almost immediately. You might look at 3-year OS and cite ~25% survival but the at-risk count has fallen from **640** → **27**, just 4% of the original sample. By the first year the sample is already down ~88%. The total number of events observed in the no-allograft arm is only about a quarter of its original size.

There's a deeper problem with that censoring too. Censoring at transplant is *informative*: the patients being removed are precisely the fitter, deeper-responding ones. So the no-allograft curve is being continuously **depleted of its own best prognostic material**. That biases it **downward**. Burnett's ~12 months should therefore be read as a **floor**, not a central estimate. It also provides a direct argument for the approach I take with Van der Maas below, which *keeps* the transplanted patients in the denominator and models their counterfactual instead of censoring them out.

Median OS in Burnett is approximately **12 months**, and landmarking (as with Kurosawa) exhibits non-monotonicity and doesn't move the median much.

5c. Van der Maas (the best comparator but the most work)

The final dataset comes from the supplementary materials of a 2025 study covering 8 consecutive HOVON-SAKK trials from 2000-2018. HOVON is the Dutch-Belgian hemato-oncology cooperative group; SAKK is the Swiss group for clinical cancer research; recruitment spanned several European countries. All eight trials shared an intensive induction backbone for newly diagnosed AML with risk-assigned post-remission therapy. It is the successor to the Breems 2005 paper.

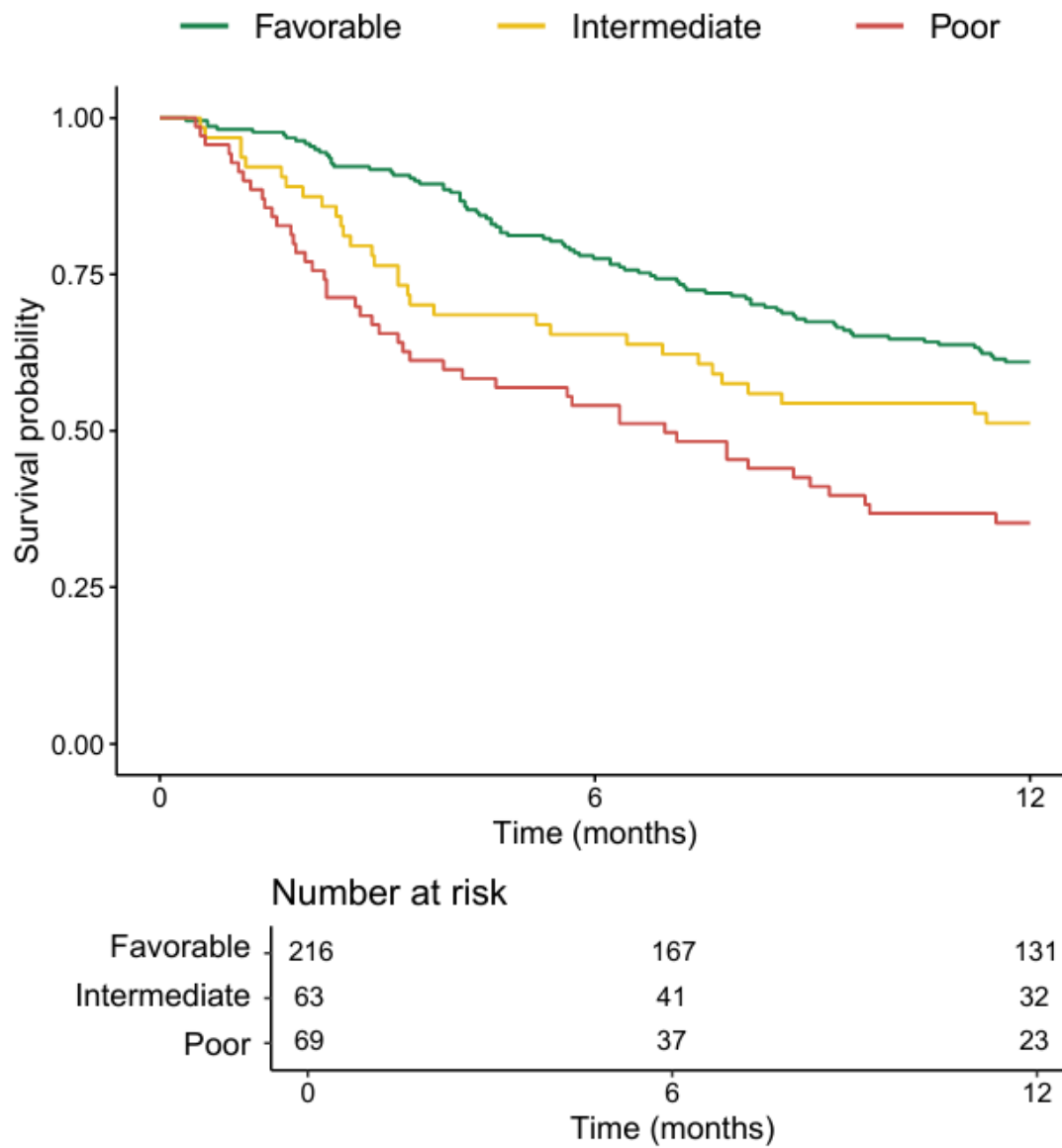
Median age at first relapse was **58 (range 18–81)**, and **348 patients achieved a second CR**.

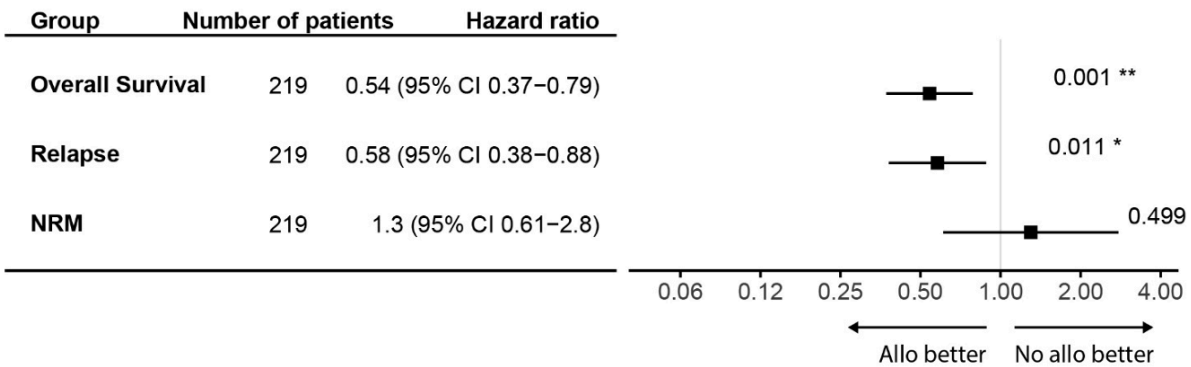
This is by far the **best comparator population** (the right age, the right era, the right geography) and it measures from **the date of CR2**. There are two problems:

1. The object it reports is **relapse-free survival (RFS)**, not OS
2. It does not separate out allograft patients (n=159)

Problem 1 is tractable because RFS is defined as "relapse or death," so **OS ≥ RFS at every point**. Therefore, RFS gives us a hard lower bound immediately, and later a convolution can be applied to get us to OS.

Problem 2 is tractable because the paper also provides a **forest plot of time-dependent allo-HCT in CR2**, comparing allo-HCT after CR2 against chemotherapy after CR2 (restricted to patients who didn't get allo-HCT before CR2 or DLI after CR2). That gives us the hazard ratios we need to reverse the transplant effect out of the curve.

C**RFS according to the revised relapse score**



Abbreviations: allo, allogeneic hematopoietic cell transplantation; NRM, non-relapse mortality, * p-value<0.05, ** p-value<0.01

One important note on the risk labels: "favorable / intermediate / poor" here are **not** ELN cytogenetic risk. They refer to the revised relapse score of the HOVON-SAKK model, which assigns more patients to its favorable group than ELN does, and the "intermediate" bar represents only the exact median score of 7 (range 1–15).

The chart runs to 12 months, where the population sits at **53.4% RFS**. Extending the curves slightly, a simple linear extrapolation puts the median RFS crossing at ~13.5 months (a cure-mixture model pushes it slightly higher).

Now I will walk through the math in sequence below (**de-transplant** → **landmark** → **convolve**).

STEP 1: DE-TRANSPLANT (recover the no-transplant relapse rate)

Everything rests on the assumed shape (as is true of all survival modelling). I assume RFS in a CR2 population takes the cure-mixture form:

$$S(t) = c + (1 - c) e^{-\lambda t}$$

This can be read as two populations stitched together. A fraction **c** are *cured* and form a flat shelf that the curve never falls below. The remaining **(1-c)** are *not cured* and relapse at rate **λ** per month, so their RFS decays as $e^{(-\lambda t)}$. At $t = 0$ everyone is event-free; at $t = \infty$ only the

cured remain, so $S \rightarrow c$. The adjustment knob is λ which represents how fast the non-cured relapse.

A useful aside: that cure fraction c is bucket 1 from Section 1. Kurosawa's no-HCT plateau sits around 18%. Burnett's no-allograft plateau sits around 25%. However, as we stated those are unreliable due to KM-instability so while we will adopt a base case here of $c = 20\%$, we will sweep across multiple c values.

Since 159 of 348 patients were transplanted after CR2, the observed (mixed) curve is the cure-mixture with the non-cured part split in two using the **0.58 relapse HR**:

$$S_{\text{mix}}(t) = c + (1 - c) \left[\underbrace{0.457 e^{-0.58\lambda t}}_{\text{transplanted}} + \underbrace{0.543 e^{-\lambda t}}_{\text{not}} \right]$$

That works as a reasonable baseline but the HR we have is **time-dependent**, so a time-dependent model is more faithful. In said model, there are two compartments: everyone starts untransplanted at hazard λ ; transplant arrives at rate κ while they remain in CR2; and only then does the relapse hazard drop to 0.58λ . Compartment A drains fast (patients leave by relapse *or* by transplant); compartment B fills from A and drains slowly. β is the gap between the two drain rates.

$$\frac{dA}{dt} = -A(\lambda + \kappa), \quad \frac{dB}{dt} = A\kappa - B(0.58\lambda)$$

Solving, with $\beta \equiv \lambda(1 - 0.58) + \kappa$:

$$A(t) = e^{-(\lambda + \kappa)t}, \quad B(t) = \frac{\kappa}{\beta} [e^{-0.58\lambda t} - e^{-(\lambda + \kappa)t}]$$

$$S_{\text{mix}}(t) = c + (1 - c)[A(t) + B(t)]$$

We know $S(12) \approx 0.534$, which lets us solve for λ at various assumed cure fractions. **At $c = 20\%$, solving simultaneously gives $\lambda \approx 0.0863/\text{month}$ and $\kappa \approx 0.0965/\text{month}$.** The fitted κ

implies a **median time-to-transplant, among those transplanted, of ≈ 2.9 months** which is squarely where real-world CR2 transplants tend to happen.

Now, we can simply **delete the transplant term** to de-transplant the curve which is shown for case **c = 20%**:

$$S_{\text{no-tx}}(t) = 0.20 + 0.80 e^{-0.0863 t}$$

Curve	Median RFS
Observed (transplant-mixed)	14.1 months
De-transplanted (no transplant)	11.4 months
De-transplant cost	-2.7 months

We can sweep across cure fractions ranging from 10-25% as well as the confidence interval of the transplant HR to get a range of **10.0-12.8 months** for our "de-transplanted" median RFS.

imposed θ	c = 0.10	c = 0.15	c = 0.20	c = 0.25
0.40	10.0	10.1	10.1	10.1
0.50	10.8	10.8	10.8	10.9
0.58	11.3	11.3	11.4	11.4
0.70	12.0	12.1	12.1	12.2
0.80	12.6	12.7	12.7	12.8

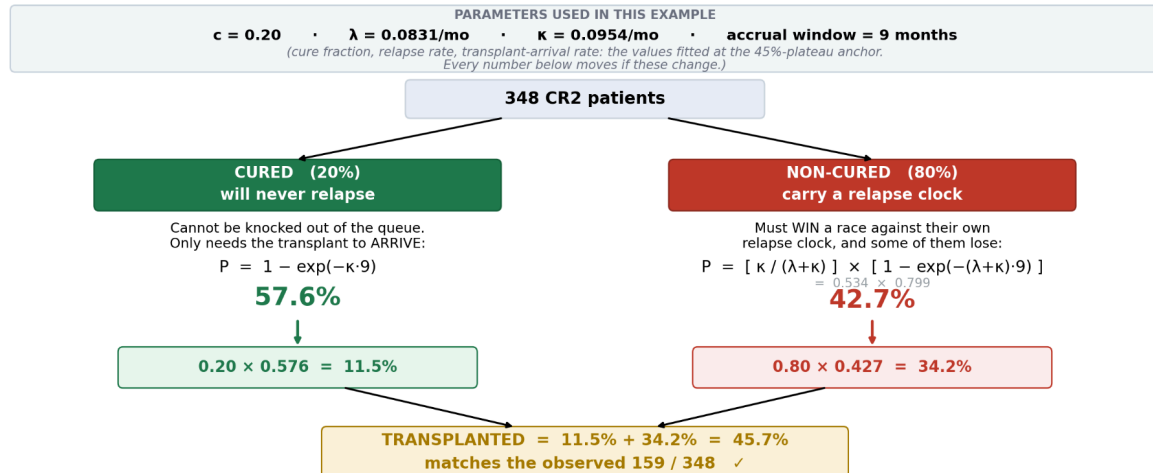
Now, a reasonable objection to the approach detailed might be that a constant hazard multiplier (0.58λ) is the wrong model, because the transplant is *curative* (i.e. it moves patients into the cured fraction **c**, raising the plateau rather than merely slowing the decay). That means the transplant effect becomes more curative over time, the HR keeps declining with follow-up, and my model **underestimates** survival past our 12-month window.

I stress-tested this by re-modelling a non-cured patient who receives a transplant as **rescued into cure with probability p**; otherwise they keep relapsing at the unchanged rate λ . This is

fully identified with three unknowns (λ , κ , p) and three anchors, the third being the long-run plateau of transplanted patients.

A worked example of rescue probability p at specific parameters

Transplant and relapse are in a race, and cured patients cannot lose it.



In this example: $11.5 / 45.7 = 25.2\%$ of recipients were already cured, vs their 20% share of the cohort

What the transplant actually accomplishes

Long-run, 45% of recipients stay relapse-free, but a quarter of them were never going to relapse anyway.

WHO GETS TRANSPLANTED



LONG-RUN OUTCOME OF THOSE SAME RECIPIENTS



0.252×1.00
already cured

+

$0.748 \times p$
rescued by transplant

=

0.45
the plateau

$p = \text{the probability a transplant converts a patient who WOULD have relapsed into one who never does.}$

$$p = (0.45 - 0.252) / 0.748$$

$p = 0.265$

roughly 1 in 4

The rescue probability p is a derived quantity. It is *lower* than the arm-level cure fraction, because some patients who receive a transplant would have been cured anyway. But it is *higher* than the assumed overall c , because of guarantee-time selection (you have to survive long enough to get transplanted at all) In the worked example below: 25.2% of recipients were already cured, 19.8% are genuinely rescued, 55% relapse despite transplant, producing the 45% plateau and $p \approx 0.265$ (roughly 1 in 4).

Sweeping the transplant plateau across 35–50% (historical allo-SCT tails) shifts the final OS-from-enrollment median by 0.1 months. The result is robust to the model class. That was the objection I most expected, and it doesn't prove problematic.

STEP 2: THE REGAL LANDMARK (the randomization window)

We can now apply the same integrated-window approach used on Kurosawa. It is actually **more appropriate here than it was there**, because we're applying it to an RFS curve: a patient who relapsed quickly (say, under six months) but wasn't yet dead would still be sitting on Kurosawa's OS curve, but would **not be eligible for REGAL**. The RFS curve removes them correctly; the OS curve does not.

At $c = 20\%$, this gives a **landmarked RFS of 12.7 months**.

τ	0	2	4	6
residual RFS median	11.4	12.1	13.1	14.3

Sensitivity: landmarked RFS median (and the gain)

imposed θ	$c = 0.10$	$c = 0.15$	$c = 0.20$	$c = 0.25$
0.40	10.5	10.9	11.4	12.4
0.50	11.2	11.6	12.1	13.1
0.58	11.7	12.1	12.7	13.6
0.70	12.5	12.9	13.4	14.3
0.80	13.0	13.4	14.0	14.9

(the gain alone)

imposed θ	$c = 0.10$	$c = 0.15$	$c = 0.20$	$c = 0.25$
0.40	+0.46	+0.82	+1.36	+2.27
0.50	+0.45	+0.80	+1.32	+2.19
0.58	+0.45	+0.79	+1.30	+2.14
0.70	+0.44	+0.78	+1.28	+2.08
0.80	+0.44	+0.77	+1.26	+2.04

STEP 3: RFS $\rightarrow$ OS (adding survival after relapse)

The adjusted curve is still relapse-free survival, which treats relapse as a terminal event. However, a patient relapsing out of CR2 does not instantly die but instead lives a while on salvage. From Section 2, we know that time in active relapse has a relatively tight range and is only a few months regardless of treatment pathway, which makes it modellable.

First, we count the ways a patient can be alive at time t :

- **a)** they're cured $\rightarrow$ contributes c
- **b)** they're non-cured but haven't relapsed yet $\rightarrow$ contributes $(1-c)e^{(-\lambda t)}$
- **c)** they're non-cured, relapsed at some earlier moment u and survived the $(t-u)$ months since $\rightarrow$ contributes the sum over all possible relapse times u :

$$\int_0^t (1 - c) \lambda e^{-\lambda u} \cdot e^{-\mu(t-u)} du$$

The first term $\lambda e^{-\lambda u}$ is the chance of relapsing right around time u ; $e^{-\mu(t-u)}$ is the chance of surviving from then to t . μ is the post-relapse death hazard, equal to $(\ln 2 / \text{post-relapse median})$ under exponential decay.

Adding (a) + (b) + (c) and resolving the integral collapses to a clean closed form:

$$S_{OS}(t) = c + (1 - c) \frac{\mu e^{-\lambda t} - \lambda e^{-\mu t}}{\mu - \lambda}$$

Solving across post-relapse medians of 3-6 months gives a **lift of roughly 5-10 months**.

It may seem strange that the lift *exceeds* the post-relapse median. However, that actually tracks. Each patient's curve is pushed out by their own delay, so the convolution adds roughly the **mean**, not the median (and for an exponential the mean is median/ $\ln 2$, i.e. ~44% larger). The lift actually lands a little above even that in this case due to a mathematical concept known as **Jensen's Inequality**.

Sensitivity: OS from enrollment ($\theta = 0.58$)

post-relapse median	c = 0.10	c = 0.15	c = 0.20	c = 0.25
3.0	16.6	17.1	17.8	18.8
4.5	19.0	19.6	20.4	21.7
6.0	21.4	22.1	23.0	24.6

post-relapse MEDIAN	its MEAN	lift @ c=0.10	lift @ c=0.15	lift @ c=0.20	lift @ c=0.25
3.0	4.33	+4.87	+4.96	+5.08	+5.24
4.5	6.49	+7.32	+7.51	+7.76	+8.13
6.0	8.66	+9.63	+9.94	+10.36	+10.98

STEP 4: STRESS-TESTING THE CHAIN (testing sensitivities to assumed hazard shapes)



The chain we have undertaken is represented above. Across the entire assumption grid (c = 0.10–0.25; HR 0.54–0.75; post-relapse median 3–6 months), OS-from-enrollment spans **15.4-25.7 months**. The majority of the sensitivity sits inside the post-relapse median assumption.

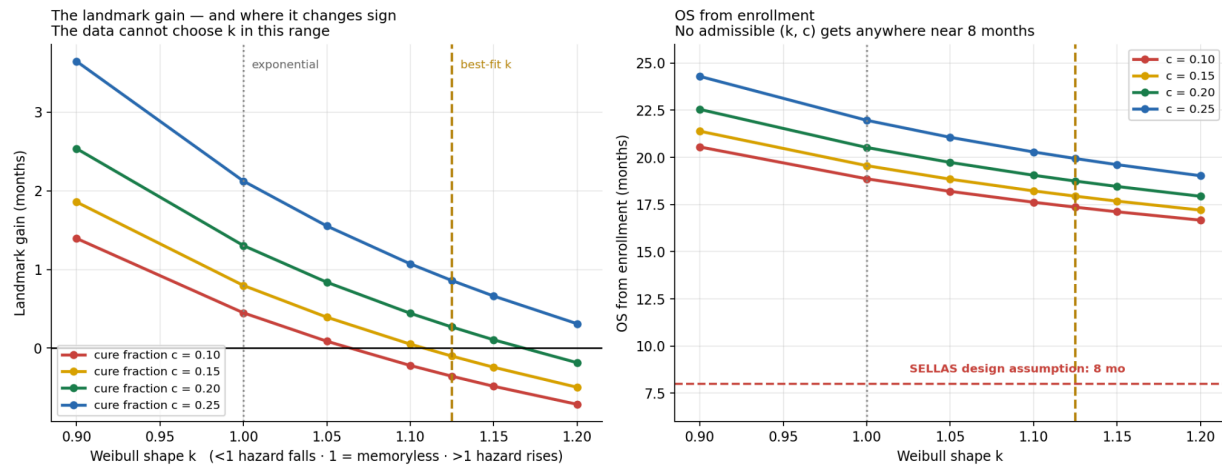
The corners

	OS from enrollment
most hostile (θ 0.40, c 0.10, pr 3 mo)	15.4
base (θ 0.58, c 0.20, pr 4.5)	20.4
most favourable (θ 0.80, c 0.25, pr 6 mo)	25.7

Note where the hostile corner lands. Even stacking every assumption against myself (the weakest cure fraction, the strongest transplant effect, the shortest post-relapse survival), the answer we arrive at is still **15.4 months**. That is roughly **double** the SELLAS design assumption. There is no admissible corner of this parameter space that gets anywhere remotely close to 8 months.

However, the result also has to be robust to the assumed *hazard shape*, not just the parameters. We will run two independent stress tests.

Stress Test #1 — the hazard shape. The first test sweeps across a range of Weibull functions that fit the data reasonably well at various cure fractions. The **landmark gain** is the step most sensitive to the Weibull fit, since it depends on extending the data past 12 months. The result holds: implied median OS varies only slightly across hazard shapes at any given parameter set. The full grid spans **14.4-27.2 months**, only modestly wider than the original 15.4-25.7.



Stress Test #2 — the post-relapse shape. This is the important one, because the RFS→OS convolution carries by far the largest influence on the result.

Recall Jensen's inequality. This is a complex mathematical topic so I won't get into the weeds but the intuition applied to our case is simple: if every patient lived *exactly* the average time after relapse, OS would just shift by that average. In reality, post-relapse survival varies from patient to patient. Some patients die sooner, some live substantially longer. Because survival curves are curved rather than perfectly linear around their median, this variability causes the median overall survival after convolution to end up slightly **greater** than simply adding the average post-relapse survival. This small difference is known as the **Jensen gap**.

$$\frac{\lambda_{\text{RFS}} \text{Var}(\text{PRS})}{2}$$

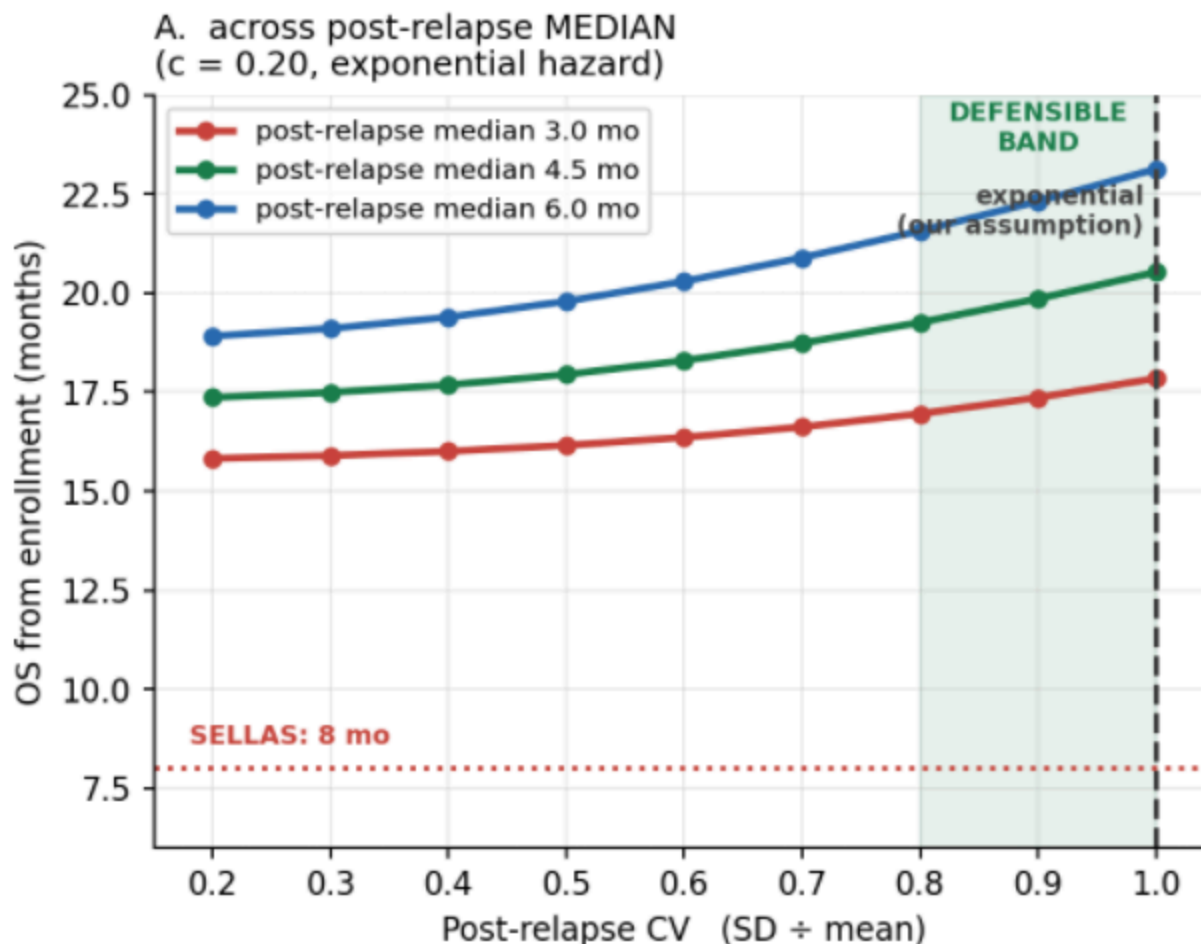
In our setting, post-relapse survival (PRS) is much shorter than relapse-free survival (RFS), so this correction is small but non-zero. More importantly, the size of the Jensen gap depends far more on **how spread out** the post-relapse survival times are than on whether they are modeled with a Weibull, exponential, gamma, or log-logistic distribution. Mathematically, the correction is driven by the variance of the added survival time and the local shape of the RFS curve around its median. In practice, this means the exact parametric family is of secondary importance; what matters most is how tightly or loosely the post-relapse survival times are clustered. Therefore,

the most natural way to stress-test this assumption is by varying the **coefficient of variation** of the post-relapse survival distribution.

	CV	meaning
deterministic	0	everyone survives exactly the same time post-relapse
Gamma shape 50	0.14	tightly clustered
Gamma shape 4	0.50	moderate spread
exponential	1.00	SD = mean

target CV	Gamma	Weibull	Lognormal
1.00	+7.56	+7.56	+7.03
0.70	+5.94	+5.85	+6.01
0.50	+5.18	+5.00	+5.32
0.30	+4.72	+4.56	+4.79

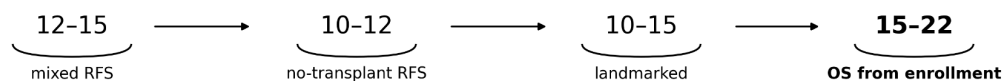
Notice that the three distribution families agree with each other far more closely than any of them varies across CV. This is exactly what the Jensen-gap formula predicts, and is the cleanest confirmation that the parametric family isn't what's driving the answer.



The key takeaway is a downward one: if post-relapse survival is more tightly clustered than exponential, I have **overestimated the RFS→OS convolution by roughly 2–3 months**, and that should come off the OS-from-enrollment figure. This is the single largest correction working *against* my estimate, and it is why I don't quote the base case of 20.4 as my answer.

STEP 5: BRINGING IT ALL TOGETHER

Sweeping across all admissible tuning knobs gives a credible band (10th-90th percentile) for the REGAL OS-from-enrollment object:



Central estimate: 18.3 months.

Now, it is worth noting that "de-transplanting" the curve has answered the question "*what if nobody had been transplanted?*" but *REGAL* will certainly see transplants, because patients need only be transplant-**ineligible at randomization**, and eligibility is a fluid condition. Cutting the other way, deliberately enrolling transplant-ineligible patients in the haplo era may mean genuinely sicker, more comorbid patients because (per Section 3), "ineligible" no longer means "unlucky." Cutting back the other way again, some of that group may be *favorably* enriched precisely because of *why* they aren't being transplanted (the CBF patient who declines transplant because their baseline odds of durable remission are already good).

It is not a perfect comparator. It is a very reasonable one.

BROAD TAKEAWAY

We have four independent lines of evidence, built on different data, in different populations, using different methods that all point towards much higher baseline survival than the number that Sellas management has been selling:

1. **Historical data.** A reasonable baseline for CR2 duration is ~7–8 months and for post-relapse survival ~4–6 months, implying **~11–14 months** as a naive baseline (~9–16 wider range). But that is a naive sum of medians, which *understates* the true blended median. Corrected, a more reasonable baseline is **~12–18 months, with >20 possible**.
2. **Kurosawa.** Median survival from CR2 onset $\approx$ **18 months** for intermediate-risk patients (but data is limited to Japan, and has an unstable tail)
3. **Burnett.** Median survival from CR2 onset $\approx$ **12 months** but in a young population subject to *adverse* selection into the no-transplant group, read off a curve whose effective sample size collapses ~88% within the first year. This is a floor, not a central estimate.
4. **Van der Maas.** The best comparator population, though it cleanly reports only RFS from CR2 rather than OS. De-transplanted, landmarked and convolved, it strongly implies **~15–22 months**, central **~18–19 months**.

My view is that the most plausible estimate for REGAL median OS in the control arm is ~16-18 months. The trial then hinges on the long-term survival tail, which will be dominated by the cytogenetic-risk composition of the patients who achieve durable remissions *and* by the transplant tail.